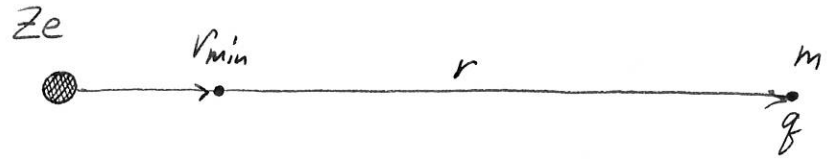
Problem 1



a) The instantaneous energy loss dE_{rad}/dt is given by Eq. (10.77)

$$\frac{dE_{rad}}{dt} = \frac{\mu_0}{6\pi c}\ddot{\vec{d}}^{\,2} = \frac{\mu_0 q^2}{6\pi c} a^2 \qquad (1)$$

The acceleration a is determined from the Newton's law

$$ma = -\frac{dU}{dr}$$

We can now transform the left-hand side of Eq. (1) as

$$\frac{dE_{rad}}{dt} = \frac{dE_{rad}}{dr}\frac{dr}{dt},$$

so that the total radiation on the approach from $r=\infty$ to $r=r_{min}$ is

$$E_{rad}^{(approach)} = \int_{\infty}^{r_{min}} \frac{dE_{rad}}{dr}\,dr = \int_{\infty}^{r_{min}} dr\, \frac{\mu_0 q^2 \left(\frac{dU}{dr}\right)^2}{6\pi c\,(dr/dt)\,m^2}$$

From the conservation of energy

$$\frac{m}{2}\left(\frac{dr}{dt}\right)^2 = \frac{mv_0^2}{2} - U(r) \Rightarrow \frac{dr}{dt} = -\sqrt{v_0^2 - \frac{2U(r)}{m}}$$

so that $$E_{rad}^{(approach)} = \frac{\mu_0 q^2}{6\pi c m^2}\int_{r_{min}}^{\infty} dr\, \frac{(dU/dr)^2}{\sqrt{v_0^2 - 2U(r)/m}}$$

Note that $$v_0^2 - \frac{2U(r_{min})}{m} = 0$$

The energy radiated during departure is equal to $E_{rad}^{(approach)}$, so that the total loss

$$E_{rad} = \frac{\mu_0 q^2}{3\pi c m^2} \int_{r_{min}}^{\infty} dr \frac{(dU/dr)^2}{\sqrt{v_0^2 - 2U(r)/m}}$$

This formula is valid as long as $E_{rad} \ll m v_0^2$.

b) $\frac{dU}{dr} = -\frac{zeq}{4\pi\varepsilon_0 r^2}$ and Eq. (2) gives

$$E_{rad} = \frac{\mu_0 q^2}{3\pi c m^2} \frac{z^2 e^2 q^2}{16\pi^2 \varepsilon_0^2} \int_{r_{min}}^{\infty} \frac{dr}{r^4 \sqrt{v_0^2 - \frac{zeq}{2\pi\varepsilon_0 m r}}} =$$

$$= \frac{\mu_0 z^2 e^2 q^4}{48\pi^3 c m^2 \varepsilon_0^2 v_0} \int_{r_{min}}^{\infty} \frac{dr}{r^{7/2}\sqrt{r - r_{min}}}$$

where $r_{min} = \frac{zeq}{2\pi\varepsilon_0 m v_0^2}$

The integral can be easily rescaled by the substitution: $r = r_{min} y$

$$\int_{r_{min}}^{\infty} \frac{dr}{r^{7/2}\sqrt{r - r_{min}}} = \frac{1}{r_{min}^3} \int_1^{\infty} \frac{dy}{y^{7/2}\sqrt{y-1}} = \frac{16}{15} \frac{1}{r_{min}^3}$$

$$E_{rad} = \frac{\mu_0 z^2 e^2 q^4}{45\pi^3 c m^2 \varepsilon_0^2 v_0} \frac{(2\pi\varepsilon_0 m v_0^2)^3}{(zeq)^3} = \frac{8\mu_0 \varepsilon_0 m v_0^5 q}{45 z e c}$$

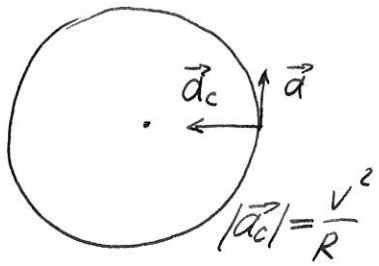
Recalling now that $\mu_0 \varepsilon_0 = \frac{1}{c^2}$ and $E_0 = \frac{m v_0^2}{2}$ we find

$$\frac{E_{rad}}{E_0} = \frac{16}{45}\left(\frac{q}{ze}\right)\left(\frac{v_0}{c}\right)^3$$

We observe that, since $v_0 \ll c$, the losses are small, $E_{rad} \ll E_0$.

Problem 2 Electric dipole radiation

$$\frac{d\mathcal{E}_d}{dt} = \frac{\mu_0 q^2 \vec{a}^2}{6\pi c}$$



The acceleration consists of a tangential component (constant in time) and a centripetal one:

$$\vec{a}^2 = a^2 + \frac{v^4}{R^2} = a^2\left(1 + \frac{a^2 t^4}{R^2}\right)$$

N revolutions take time t_N to perform:

$$\frac{a t_N^2}{2} = 2\pi R N \implies t_N = \sqrt{\frac{4\pi R N}{a}}$$

The amount of energy radiated from $t=0$ to $t=t_N$:

$$\mathcal{E}_N^{(d)} = \frac{\mu_0 q^2}{6\pi c}\int_0^{t_N} \vec{a}^2(t)\,dt = \frac{\mu_0 q^2 a^2}{6\pi c}\int_0^{t_N} dt\left(1 + \frac{a^2 t^4}{R^2}\right) =$$

$$= \frac{\mu_0 q^2 a^2}{6\pi c}\left(t_N + \frac{a^2 t_N^5}{R^2}\right) = \frac{\mu_0 q^2 a^2}{6\pi c}\sqrt{\frac{4\pi R N}{a}}\left(1 + \frac{16\pi^2 R^2 N^2 a^2}{a^2 R^2}\right)$$

$$= \frac{\mu_0 q^2 a^{3/2}\sqrt{RN}}{3\sqrt{\pi}\, c}\left(1 + 16\pi^2 N^2\right)$$

Magnetic dipole radiation

$$\frac{d\mathcal{E}_m}{dt} = \frac{\mu_0}{6\pi c^3}\ddot{\vec{m}}^2 \quad , \text{ where } \quad \vec{m} = \frac{q}{2}\vec{r}\times\vec{v}$$

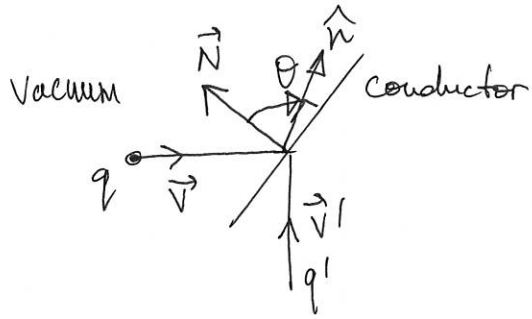
$$\vec{r} = R\hat{x}\cos\frac{at^2}{2R} + R\hat{y}\sin\frac{at^2}{2R}$$

$$\vec{v} = \dot{\vec{r}} = -\hat{x}\,at\sin\frac{at^2}{2R} + \hat{y}\,at\cos\frac{at^2}{2R}$$

$$\vec{r}\times\vec{v} = Rat\hat{z}\cos^2\left(\frac{at}{2R}\right) + Rat\hat{z}\sin^2\left(\frac{at}{2R}\right) = Rat\hat{z}$$

Since m ~ t, we have that d^2(m)/dt^2 = 0 and magnetic dipole radiation is absent.

(P3) Transition radiation spectral intensity.



Observe that in this problem we integrate only over vacuum region
$\Rightarrow \theta \in [0, \pi/2]$

Non-relativistic limit: $|\hat{n} \times \vec{N}| = \sin\theta$

$$\frac{d\mathcal{E}}{d\omega} = \frac{\mu_0 q^2}{16\pi^3 c} \int d\Omega \, |\hat{n} \times (\vec{v} - \vec{v}')|^2 = \frac{\mu_0 q^2}{4\pi^3 c} (\vec{v} \cdot \vec{N})^2 \cdot 2\pi$$

$$\cdot \underbrace{\int_0^{\pi/2} d\theta \sin\theta \, \sin^2\theta}_{x = \cos\theta} = \frac{\mu_0 q^2}{3\pi^2 c} (\vec{v} \cdot \vec{N})^2$$

$$\int_0^1 dx (1 - x^2) = 1 - \frac{1}{3} = \frac{2}{3}$$

Relativistic limit:

$$\frac{d\mathcal{E}}{d\omega} = \frac{\mu_0 q^2}{16\pi^3 c} \int d\Omega \left(\frac{\hat{n} \times \vec{v}'}{1 - \frac{1}{c} \hat{n} \cdot \vec{v}'} \right)^2 = \frac{\mu_0 q^2}{16\pi^3 c} \cdot 2\pi \cdot \int_0^{\theta_0} d\theta \sin\theta$$

$$\cdot \left(\frac{2c\gamma^2 \theta}{1 + \gamma^2 \theta^2} \right)^2$$

where we used small angle approximation (just like in the Lecture).

We know that ~~most~~ radiation is emitted within a cone of angle $\theta \sim \frac{1}{\gamma} \ll 1$

$\Rightarrow$ θ- integral can be approximated as

$$I = \int_0^{\theta_0} d\theta \, \theta \, \frac{\theta^2}{(1 + \gamma^2 \theta^2)^2}$$

where $\theta_0 \approx 1$ is ~~an~~ an upper cut-off.

Then $I = \frac{1}{2}\int_0^{y_0} dy \, \frac{y}{(1+\gamma^2 y)^2} = -\frac{1}{2\gamma^2}\int \cancel{dy}\left(\frac{1}{1+\gamma^2 y}\right) y =$

$y = \theta^2, \; y_0 = \theta_0^2$

$$= -\frac{1}{2\gamma^2}\left\{ \frac{y}{1+\gamma^2 y}\Big|_0^{y_0} - \int_0^{y_0} \frac{dy}{1+\gamma^2 y} \right\} =$$

$$= -\frac{1}{2\gamma^2}\left\{ \frac{y_0}{1+\gamma^2 y_0} - \frac{1}{\gamma^2}\ln(1+\gamma^2 y)\Big|_0^{y_0} \right\} =$$

$$= -\frac{1}{2\gamma^2}\frac{y_0}{1+\gamma^2 y_0} + \frac{1}{2\gamma^4}\ln(1+\gamma^2 y_0)$$

$$\Rightarrow \frac{d\mathcal{E}}{d\omega} = \frac{\mu_0 q^2}{16\pi^3 c}\cdot 2\pi \cdot 4c^2\gamma^4 \cdot \frac{1}{2\gamma^4}\left\{ \ln(1+\gamma^2 y_0) - \frac{\gamma^2 y_0}{1+\gamma^2 y_0} \right\}$$

$$= \frac{\mu_0 q^2 c}{4\pi^2}\left\{ \ln\left[\gamma^2(\gamma^{-2}+y_0)\right] - \frac{\gamma^2 y_0}{1+\gamma^2 y_0} \right\}$$

In the $\gamma \to \infty$ limit (which is $v \to c$ limit)

$$\ln\gamma^2 \gg \ln\left(y_0 + \frac{1}{\gamma^2}\right) = \ln(y_0) \text{ and}$$

$$\ln\gamma^2 \gg \frac{\gamma^2 y_0}{1+\gamma^2 y_0} \xrightarrow[\gamma\to\infty]{} 1 .$$

$\Rightarrow$ with such logarithmic accuracy we get

$$\frac{d\mathcal{E}}{d\omega} = \frac{\mu_0 q^2 c}{2\pi^2}\ln(\gamma).$$

This can also be derived by full integration over θ.